Math 534H - Spring 2026 - Lecture 25

Q: How about representing functions ϕ using the full trigonometric system $\left\{1, \cos\left(\frac{n\pi x}{\ell}\right), \sin\left(\frac{n\pi x}{\ell}\right)\right\}$

↳ FULL FOURIER SERIES

That is

$$\phi(x) \overset{?}{=} \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{\ell}\right) + B_n \sin\left(\frac{n\pi x}{\ell}\right)$$

where A_0, A_n, B_n, $n \geq 1$ can be computed using the integrals in Lecture 23

3, 4, 5, 6, 7. But we need to work

now on a full symmetric interval $(-\ell, \ell)$

Now

$$A_0 = \frac{1}{\ell} \int_{-\ell}^{\ell} \phi(x)\, dx$$

$$A_n = \frac{1}{\ell} \int_{-\ell}^{\ell} \phi(x) \cos\left(\frac{n\pi x}{\ell}\right) ; \quad n \geq 1$$

$$B_n = \frac{1}{\ell} \int_{-\ell}^{\ell} \phi(x) \sin\left(\frac{n\pi x}{\ell}\right) , \quad n \geq 1$$

To understand this further :

<u>Definition</u> 1 An even function ϕ is a function such that $\phi(-x) = \phi(x)$

No conditions on ϕ at 0

However if you want ϕ to be - say differentiable with a continuous derivative at 0, then you need to have

$$\phi'(0) = 0 .$$

2 An ODD function is a function ϕ /

$\phi(-x) = -\phi(x)$. Note that if ϕ is ODD then $\phi(0) = 0$.

In the Definitions above ϕ could be defined on all of $\mathbb{R}$ or on $\underbrace{(-\ell, \ell)}_{\text{symmetric around } 0}$.

PROPERTIES:

- Product of ODD = EVEN
- Product of even = Even
- Product of $\left.\begin{pmatrix} \text{ODD} \cdot \text{Even} \\ \text{Even} \cdot \text{Odd} \end{pmatrix}\right\} \to$ ODD.
- SUM OF TWO ODD functions is ODD

- **SUM OF two EVEN functions is EVEN**

BUT (!) : Sum of odd and even could be anything !

- In fact given any function $f(x)$ we can always write it as the sum of an EVEN and ODD function as follows :

$$\boxed{f(x) = \underbrace{\frac{1}{2}\left[f(x)+f(-x)\right]}_{\text{EVEN}} + \underbrace{\frac{1}{2}\left[f(x)-f(-x)\right]}_{\text{ODD}}}$$

Remarks:

1. The $\sin\left(\frac{n\pi x}{\ell}\right)$ is ODD and periodic with period 2ℓ as a function of $x \in \mathbb{R}$. So the SINE series – IF IT CONVERGES – is also ODD and periodic of period 2ℓ.

So the sine series can be regarded as an expansion of a ("nice") function on $\mathbb{R}$ that is ODD and periodic with period 2ℓ.

2. The $\cos\left(\frac{n\pi x}{\ell}\right)$ IS EVEN and periodic with period 2ℓ as a function of $x \in \mathbb{R}$. So the

COSINE series – IF IT CONVERGES – is also an EVEN and periodic function of period 2ℓ.

So the cosine series can be regarded as an expansion of a ("nice") function on $\mathbb{R}$ that is EVEN and periodic with period 2ℓ.

REMARK: Note below we are integrating over $(-\ell, \ell)$ = a symmetric interval around 0.

$$\int_{-\ell}^{\ell} (\text{ODD})\, dx = 0 \quad ; \quad \int_{-\ell}^{\ell} (\text{EVEN})\, dx = 2\int_{0}^{\ell} (\text{EVEN})\, dx$$

• EVEN AND ODD EXTENSIONS

How do we extend a function defined on $(0, \ell)$ in a unique fashion to be either EVEN or ODD function on the interval $(-\ell, \ell)$

$$\phi_{EVEN}(x) = \begin{cases} \phi(x) & 0 < x < \ell \\ \phi(-x) & -\ell < x < 0 \end{cases}$$

• NOTE: There is no condition that needs to be imposed at $x=0$. If ϕ is continuous then ϕ_{even} can be made to be continuous at $x=0$ by defining $\phi_{even}(0)$ to be $\lim_{x \to 0^+} \phi(x)$ $\left(= \lim_{x \to 0^-} \phi(-x)\right)$.

$$\phi_{ODD}(x) := \begin{cases} \phi(x) & 0 < x < \ell \\ 0 & x = 0 \\ -\phi(-x) & -\ell < x < 0 \end{cases}$$

<u>NOTE</u>: Now one needs to specify the value of ϕ_{ODD} at $x = 0$. However note that ϕ_{ODD} does not need to be continuous at $x = 0$ even if ϕ is.

Consider $\phi(x) = 1$ on $(0, \ell)$. Then

$$\phi_{ODD}(x) = \begin{cases} 1 & 0 < x < \ell \\ 0 & x = 0 \\ -1 & -\ell < x < 0 \end{cases}$$

<u>Remark</u> On the interval $(0, \ell)$ if we have wave or heat/diffusion equation <u>with</u>:

- zero Dirichlet BC, then we can view these on the interval $(-\ell, \ell)$ by doing an ODD extension.
- zero Neumann BC, then we can view these on the interval $(-\ell, \ell)$ by doing an EVEN extension.

<u>DEFINITION</u> On the interval $(-\ell, \ell)$ we say that we have PERIODIC BOUNDARY conditions for a solution $u(x,t)$ if

$u(-\ell, t) = u(\ell, t)$. (one could also specify instead that $u_x(-\ell, t) = u_x(\ell, t)$).

Periodic BC correspond then to the periodic extension to $\mathbb{R}$ (that is can extend as a periodic function from $(-\ell, \ell)$ to $\mathbb{R}$).

Coming back to page 1 of this lecture let us see an example of a FULL FOURIER series expansion. Consider the function

$\phi(x) = x$ on $(-1, 1)$. Recall in Lecture 24 we also considered $\underline{\phi(x) = x}$ but in the interval $(0,1)$ and computed both its cosine series and its sine series. Now we are in $(-1,1)$ and want to consider its FULL Fourier Series as in page 1 of this lecture. That is:

We want to write x as

$$\frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos(n\pi x) + B_n \sin(n\pi x)$$

$(\ell = 1)$.

Then according to what we learned we need to compute:

$$A_0 = 1. \int_{-1}^{1} x \, dx = \frac{1}{2} x^2 \Big|_{x=-1}^{x=1} = 0 . \qquad (\ell = 1)$$

(also note $\phi(x) = x$ is ODD so $\int_{-1}^{1} \text{odd } dx = 0$).

Next for any $m \geq 1$:

$$A_m = 1. \int_{-1}^{1} x \cos(m\pi x) \, dx \underset{\text{IBP}}{=} \frac{1}{\pi^2 m^2} \Big[\underbrace{\cos(m\pi) - \cos(-m\pi)}_{=0} \Big]$$

$$= 0 .$$

$$B_m = 1 \int_{-1}^{1} x \sin(m\pi x) \, dx \underset{\text{IBP}}{=} - \frac{1}{m\pi} \Big[\cos(m\pi) + \cos(-m\pi) \Big]$$

$$= (-1)^{m+1} \frac{2}{m\pi} \Longrightarrow$$

All in all we get that the Full Fourier series for

$\phi(x) = x$ on $(-1, 1)$ is :

$$\frac{2}{\pi}\sin(\pi x) - \frac{1}{\pi}\sin(2\pi x) + \frac{2}{3\pi}\sin(3\pi x) - \frac{1}{2\pi}\sin(4\pi x) + \cdots$$

That is on $(-1, 1)$

$$x \sim 2 \cdot \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n\pi} \sin(n\pi x).$$

Note $\phi(x) = x$ is ODD on $(-1, 1)$ so we expected its full Fourier series would be reduced to a sine series but one that now is in $(-\ell, \ell)$.